



ASoC DAPM



- ▶ DAPM stands for Dynamic Audio Power Management.
- ▶ The goal is to save as much power as possible by shutting down audio routes that are not in use.
- ▶ This may affect the whole card or just part of it.
- ▶ To achieve this, the topology needs to be described. For this we have two objects: DAPM widgets and DAPM routes.
- ▶ The DAPM widgets represent various components of an audio system, such as audio inputs, outputs, mixers, and amplifiers.
- ▶ The routes are connecting widgets together.



snd_soc_dapm_widget

- ▶ An array of `struct snd_soc_dapm_widget` is registered by the component.
- ▶ Many helpers exist to avoid filling the struct manually:

`include/sound/soc-dapm.h`

```
#define SND_SOC_DAPM_INPUT(wname)
{
    .id = snd_soc_dapm_input, .name = wname, .kcontrol_news = NULL,
    .num_kcontrols = 0, .reg = SND_SOC_NOPM }

#define SND_SOC_DAPM_OUTPUT(wname)
{
    .id = snd_soc_dapm_output, .name = wname, .kcontrol_news = NULL,
    .num_kcontrols = 0, .reg = SND_SOC_NOPM }

#define SND_SOC_DAPM_MIC(wname, wevent)
{
    .id = snd_soc_dapm_mic, .name = wname, .kcontrol_news = NULL,
    .num_kcontrols = 0, .reg = SND_SOC_NOPM, .event = wevent,
    .event_flags = SND_SOC_DAPM_PRE_PMU | SND_SOC_DAPM_POST_PMD}

[...]

#define SND_SOC_DAPM_PGA(wname, wreg, wshift, winvert,
    wcontrols, wncontrols)
{
    .id = snd_soc_dapm_pga, .name = wname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),
    .kcontrol_news = wcontrols, .num_kcontrols = wncontrols}

[...]

#define SND_SOC_DAPM_MUX(wname, wreg, wshift, winvert, wcontrols)
{
    .id = snd_soc_dapm_mux, .name = wname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),
```



snd_soc_dapm_widget

```
.kcontrol_news = wcontrols, .num_kcontrols = 1}  
#define SND_SOC_DAPM_DEMUX(wname, wreg, wshift, winvert, wcontrols)  
{  
    .id = snd_soc_dapm_demux, .name = wname,  
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),  
    .kcontrol_news = wcontrols, .num_kcontrols = 1}
```



snd_soc_dapm_widget

include/sound/soc-dapm.h

```
#define SND_SOC_DAPM_DAC(wname, stname, wreg, wshift, winvert)
{
    .id = snd_soc_dapm_dac, .name = wname, .sname = stname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert) }

#define SND_SOC_DAPM_DAC_E(wname, stname, wreg, wshift, winvert,
                           wevent, wflags)
{
    .id = snd_soc_dapm_dac, .name = wname, .sname = stname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),
    .event = wevent, .event_flags = wflags}

#define SND_SOC_DAPM_ADC(wname, stname, wreg, wshift, winvert)
{
    .id = snd_soc_dapm_adc, .name = wname, .sname = stname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert), }

#define SND_SOC_DAPM_ADC_E(wname, stname, wreg, wshift, winvert,
                           wevent, wflags)
{
    .id = snd_soc_dapm_adc, .name = wname, .sname = stname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),
    .event = wevent, .event_flags = wflags}

/* generic widgets */
#define SND_SOC_DAPM_REG(wid, wname, wreg, wshift, wmask, won_val, woff_val)
{
    .id = wid, .name = wname, .kcontrol_news = NULL, .num_kcontrols = 0,
    .reg = wreg, .shift = wshift, .mask = wmask,
    .on_val = won_val, .off_val = woff_val, }
```



snd_soc_dapm_widget

```
#define SND_SOC_DAPM_SUPPLY(wname, wreg, wshift, winvert, wevent, wflags)
{
    .id = snd_soc_dapm_supply, .name = wname,
    SND_SOC_DAPM_INIT_REG_VAL(wreg, wshift, winvert),
    .event = wevent, .event_flags = wflags}

#define SND_SOC_DAPM_REGULATOR_SUPPLY(wname, wdelay, wflags)
{
    .id = snd_soc_dapm_regulator_supply, .name = wname,
    .reg = SND_SOC_NOPM, .shift = wdelay, .event = dapm_regulator_event,
    .event_flags = SND_SOC_DAPM_PRE_PMU | SND_SOC_DAPM_POST_PMD,
    .on_val = wflags}
```



DAPM example

sound/soc/codecs/pcm3168a.c

```
static const struct snd_soc_dapm_widget pcm3168a_dapm_widgets[] = {
    SND_SOC_DAPM_DAC("DAC1", "Playback", PCM3168A_DAC_OP_FLT,
        PCM3168A_DAC_OPEDA_SHIFT, 1),
    SND_SOC_DAPM_DAC("DAC2", "Playback", PCM3168A_DAC_OP_FLT,
        PCM3168A_DAC_OPEDA_SHIFT + 1, 1),
    SND_SOC_DAPM_DAC("DAC3", "Playback", PCM3168A_DAC_OP_FLT,
        PCM3168A_DAC_OPEDA_SHIFT + 2, 1),
    SND_SOC_DAPM_DAC("DAC4", "Playback", PCM3168A_DAC_OP_FLT,
        PCM3168A_DAC_OPEDA_SHIFT + 3, 1),

    SND_SOC_DAPM_OUTPUT("AOUT1L"),
    SND_SOC_DAPM_OUTPUT("AOUT1R"),
    SND_SOC_DAPM_OUTPUT("AOUT2L"),
    SND_SOC_DAPM_OUTPUT("AOUT2R"),
    SND_SOC_DAPM_OUTPUT("AOUT3L"),
    SND_SOC_DAPM_OUTPUT("AOUT3R"),
    SND_SOC_DAPM_OUTPUT("AOUT4L"),
    SND_SOC_DAPM_OUTPUT("AOUT4R"),
}
```

Note: on the **PCM3168A** DACs and ADCs can only be powered down in pairs.



DAPM example

sound/soc/codecs/pcm3168a.c

```
SND_SOC_DAPM_ADC("ADC1", "Capture", PCM3168A_ADC_PWR_HPFB,  
                 PCM3168A_ADC_PSVAD_SHIFT, 1),  
SND_SOC_DAPM_ADC("ADC2", "Capture", PCM3168A_ADC_PWR_HPFB,  
                 PCM3168A_ADC_PSVAD_SHIFT + 1, 1),  
SND_SOC_DAPM_ADC("ADC3", "Capture", PCM3168A_ADC_PWR_HPFB,  
                 PCM3168A_ADC_PSVAD_SHIFT + 2, 1),
```

```
SND_SOC_DAPM_INPUT("AIN1L"),  
SND_SOC_DAPM_INPUT("AIN1R"),  
SND_SOC_DAPM_INPUT("AIN2L"),  
SND_SOC_DAPM_INPUT("AIN2R"),  
SND_SOC_DAPM_INPUT("AIN3L"),  
SND_SOC_DAPM_INPUT("AIN3R")
```

```
};
```




- ▶ An array of `struct snd_soc_dapm_route` is registered by the component to define the routes.

`include/sound/soc-dapm.h`

```
struct snd_soc_dapm_route {
    const char *sink;
    const char *control;
    const char *source;

    /* Note: currently only supported for links where source is a supply */
    int (*connected)(struct snd_soc_dapm_widget *source,
                    struct snd_soc_dapm_widget *sink);

    struct snd_soc_dobj dobj;
};
```



DAPM routes example

sound/soc/codecs/pcm3168a.c

```
static const struct snd_soc_dapm_route pcm3168a_dapm_routes[] = {
    /* Playback */
    { "AOUT1L", NULL, "DAC1" },
    { "AOUT1R", NULL, "DAC1" },

    { "AOUT2L", NULL, "DAC2" },
    { "AOUT2R", NULL, "DAC2" },

    { "AOUT3L", NULL, "DAC3" },
    { "AOUT3R", NULL, "DAC3" },

    { "AOUT4L", NULL, "DAC4" },
    { "AOUT4R", NULL, "DAC4" },

    /* Capture */
    { "ADC1", NULL, "AIN1L" },
    { "ADC1", NULL, "AIN1R" },

    { "ADC2", NULL, "AIN2L" },
    { "ADC2", NULL, "AIN2R" },

    { "ADC3", NULL, "AIN3L" },
    { "ADC3", NULL, "AIN3R" }
};
```